

What test statistic do I use?

Name	Use	Parameter/Statistic	Hypotheses (See note at bottom)	Assumptions	Test Statistic	Confidence Interval
One proportion z-test	Testing whether a <i>discrete</i> (binary) outcome is different from neutral/the status quo. <u>Examples</u> - Does a new drug change the prevalence of a disease (a.k.a. the rate at which a disease occurs)? - Does a new manufacturing machine reduce the probability of outputting a defective product? - Have marriage rates changed from 50 years ago?	Population parameter: p (proportion of times we expect an event to occur) Sample statistic: $\hat{p} = \frac{x}{n}$ (number of times event occurred, divided by number of trials observed)	H ₀ : There is no difference between the actual rate of Event X and what we expect under status quo. $H_0: p = p_0$ Ha: The actual rate for event X is different from/higher than/lower than what we expect under status quo. $H_a: p \neq p_0$ (or >, or <)	• Random sample • Large sample size: $np_0 > 10$ and $n(1 - p_0) > 10$ • Large population: the population is at least 10 times bigger than the sample • Observations (trials) are independent	$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1 - p_0)}{n}}}$	$\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$
Two proportion z-test	Comparing a <i>discrete (binary)</i> outcome between two different groups. In other words, a binary explanatory variable and a binary response. <u>Examples</u> - Do patients who received a real flu vaccine have a lower chance of getting the flu, relative to patients who got a placebo (fake) vaccine? - Are people who see one online advertisement more likely to click on the ad the people who see a different online advertisement?	Population parameters: p_1, p_2 Sample statistics: $\hat{p}_1, \hat{p}_2$ Pooled sample proportion: $\hat{p} = \frac{n_1\hat{p}_1 + n_2\hat{p}_2}{n_1 + n_2}$ n_1 is the number of trials in group 1 and n_2 is the number of trials in group 2	H ₀ : There is no difference between the rate of Event X for group 1 and group 2. $H_0: p_1 = p_2$ Ha: The actual mean of some population measurement is different from/higher than/lower than what we expect under status quo. $H_a: p_1 \neq p_2$ (or >, or <)	• Random samples • Large sample sizes: $n_1\hat{p} > 10$ and $n_1(1 - \hat{p}) > 10$ $n_2\hat{p} > 10$ and $n_2(1 - \hat{p}) > 10$ • People were randomly assigned to each group • Observations within each group are independent	$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1 - \hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$	$(\hat{p}_1 - \hat{p}_2) \pm z^* \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}}$
One sample t-test	Testing whether a <i>continuous</i> outcome is different from the neutral/status quo. <u>Examples</u> - Do statistics students at UCLA spend more time studying than a typical UCLA student? - Does a new drug reduce blood pressure in patients?	Population parameter: μ (true population mean for some measurement) Sample statistic: $\bar{x} = \frac{\sum x}{n}$ (observed mean for some given sample)	H ₀ : There is no difference between the actual mean of some population measurement and what we expect under status quo. $H_0: \mu = \mu_0$ Ha: The actual mean of some population measurement is different from/higher than/lower than what we expect under status quo. $H_a: \mu \neq \mu_0$ (or >, or <)	• Random sample • Large sample size: $n > 25$ • Large population: the population is at least 10 times bigger than the sample. • Observations are independent	$t = \frac{\bar{x} - \mu_0}{\sqrt{\frac{s^2}{n}}}$ Degrees of freedom = $n - 1$	$\bar{x} \pm t_{\frac{\alpha}{2}, df}^* \sqrt{\frac{s^2}{n}}$
Unpaired two sample t-test	Comparing a <i>continuous</i> outcome between two different groups. In other words, a binary explanatory variable and a continuous response. <u>Examples</u> - Is there a difference in final exam scores for students with a tutor and students without a tutor? - Are men taller than women? - Do two species of elephants eat the same weight of food per day? - Do art critics assign higher monetary values to paintings by male artists than those by female artists?	Population parameters: μ_1, μ_2 Sample statistics: $\bar{x}_1, \bar{x}_2$ n_1 is the number of observations in group 1 and n_2 is the number of observations in group 2	H ₀ : There is no difference in the mean response for group 1 and group 2. $H_0: \mu_1 = \mu_2$ Ha: The actual mean of some population measurement is different from/higher than/lower than what we expect under status quo. $H_a: \mu_1 \neq \mu_2$ (or >, or <)	• Random samples • Large sample size: $n_1 > 25, n_2 > 25$ • Observations are independent between samples and within samples	$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$ Degrees of freedom = $n_1 + n_2 - 2$	$(\bar{x}_1 - \bar{x}_2) \pm t^* \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$
Paired two sample t-test	Comparing a <i>continuous</i> outcome between two different groups, when the observations in the two groups are pairs. Related to the one sample t-test <u>Examples</u> - Do people who smoke cigarettes have different heart rates from those who do not smoke? A study might compare the heart rates of several sets of twins – one twin smokes and the other does not. - Are husbands taller than their wives? - Do people earn more money after taking a job training course? (The same set of people is observed twice – once before and once after the job training course.)	Population parameters: μ_{diff} Sample statistic: $\bar{x}_{diff} = \frac{\sum_i (x_{i1} - x_{i2})}{n}$ Group 1 and group 2 must have the same sample size, which we'll call n . The i refers to each individual pair in n	H ₀ : There is no difference in the mean response for group 1 and group 2. $H_0: \mu_{diff} = 0$ Ha: The actual mean of some population measurement is different from/higher than/lower than what we expect under status quo. $H_a: \mu_{diff} \neq 0$ (or >, or <)	• Random sample of differences (or pairs) • Large sample size (number of pairs): $n > 25$ • Differences (or the pairs) are independent	$t = \frac{\bar{x}_{diff}}{\sqrt{\frac{s_{diff}^2}{n}}}$ Degrees of freedom = $n - 1$	$\bar{x}_{diff} \pm t^* \sqrt{\frac{s_{diff}^2}{n}}$

Adapted from Section 8.1 of *Essential Statistics* (Ryan, Wong, and Gould):

- **The null hypothesis (H_0)** is the neutral, status quo, skeptical statement **about the population parameter**. In the context of researching new ideas, the null hypothesis will always have an = (equals) sign.
- **The alternative hypothesis (H_a)** is the research hypothesis. It is a statement about the value of a parameter that we intend to demonstrate is true. The alternative hypothesis has either a $\neq$ (not equal), $>$ (greater than), or $<$ (less than) sign. **The sign depends on the context of the question.**

What is the difference between a hypothesis test and a confidence interval?

- To predict a range for the “true” value, compute the confidence interval.
- To test whether a claim (hypothesis) about the “true” value is plausible, use a hypothesis testing. Compute the test statistic and get the p-value.

Steps for a hypothesis test:

- 1) State your null and alternate hypotheses. (Replace p_0 or μ_0 with the correct value from the question.)
- 2) Check your assumptions – can you assume that your sample is random? Is your sample large enough?
- 3) Calculate your sample statistic and get your p-value.
- 4) Make a decision to reject or fail to reject your null hypothesis (by comparing your p-value to alpha).